

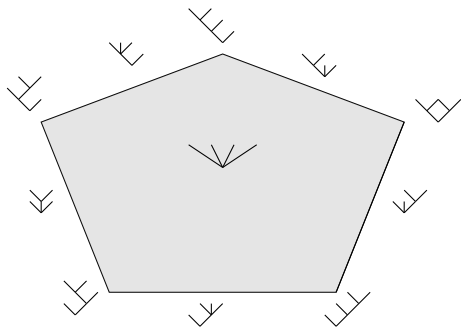
CATEGORY THEORY WITHIN A 2-CATEGORY

Sophie d’Espalungue

Séminaire Itinérant de Catégories

June, 2025

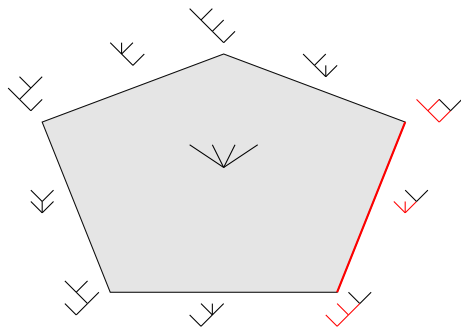
THE FACE POSETS OF THE ASSOCIAHEDRA



$$\text{pentagon}_4 \in \text{TOP}$$

TOP	pentagon_r	Associahedra
CAT	$\text{pentagon}(r)$	Poset of faces ordered by inclusion

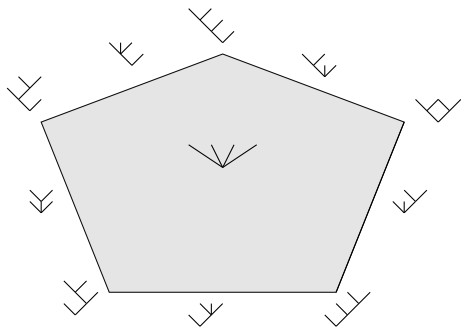
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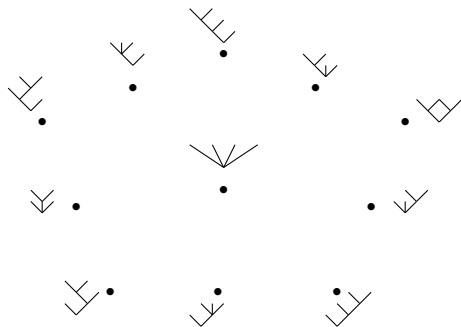
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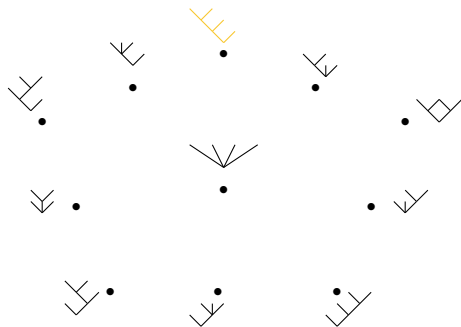
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$$\Diamond(4) \in \text{CAT}$$

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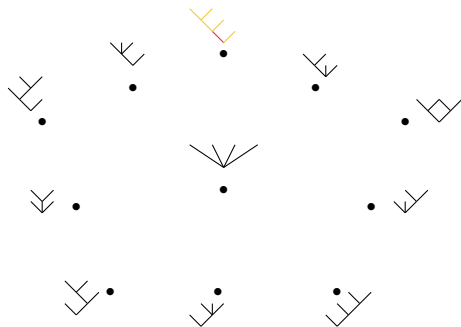
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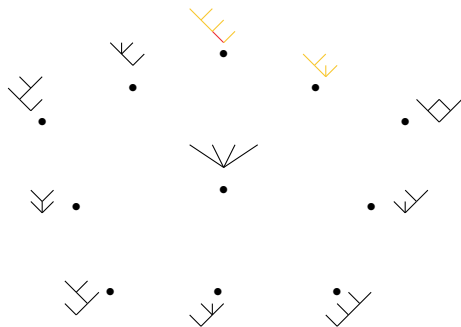
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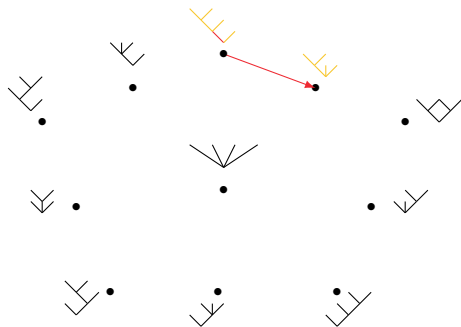
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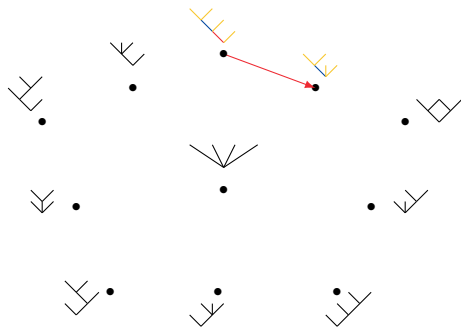
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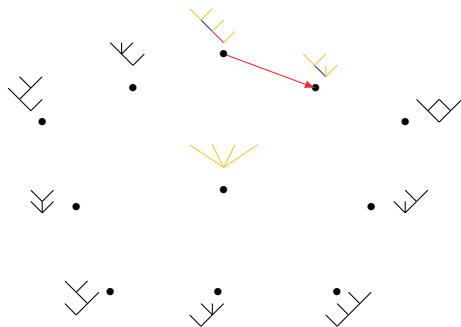
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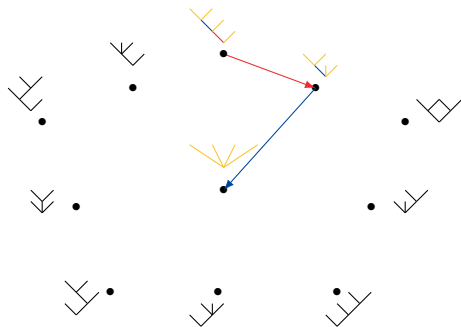
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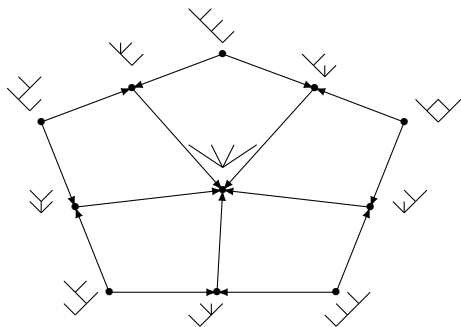
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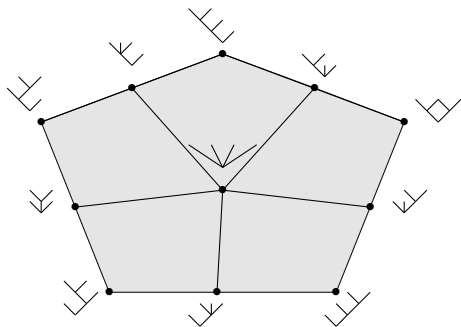
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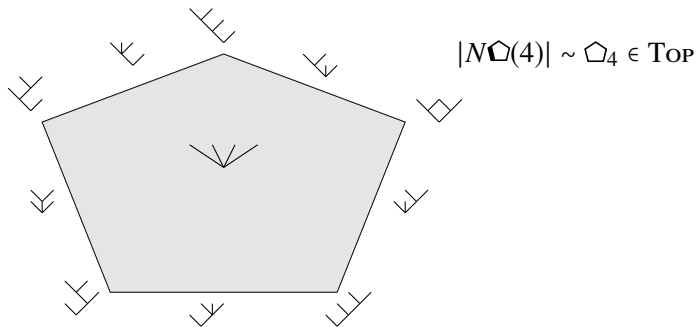
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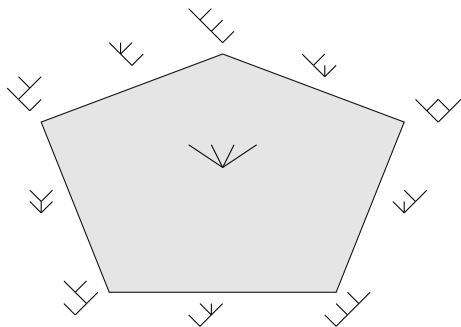
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THE FACE POSETS OF THE ASSOCIAHEDRA



OP_{TOP}		Associahedra
OP_{CAT}		Face poset operad (trees and edges contractions)

SLIGHTLY SHIFTING PERSPECTIVE

THE ASSOCIAHEDRA AS A MORPHISM OF CATEGORICAL OPERADS

$$\text{pentagon} \bullet \in \text{OP}_{\text{TOP}}^{\mathbb{N}} := \text{MON}_{(\text{CAT}, \times)} \left(\text{TOP}^{\mathbb{N}}, \circ \right)$$

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Equivalently,

$$r \in \mathbb{N} \quad \Rightarrow \quad \text{Pent}_r : * \rightarrow \text{TOP} \quad \text{in CAT}$$

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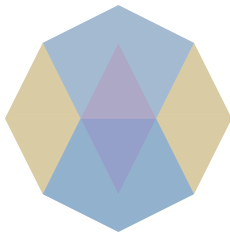
We get

$$\text{Pent} : \text{Pent} \rightarrow \mathbf{TOP} \quad \text{in OP}_{\text{CAT}}^{\mathbb{N}}$$

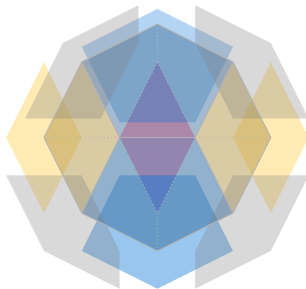
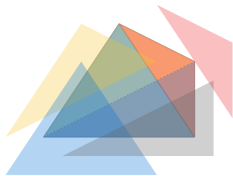
CAN WE BUILD A NERVE OUT OF THE ASSOCIAHEDRA?



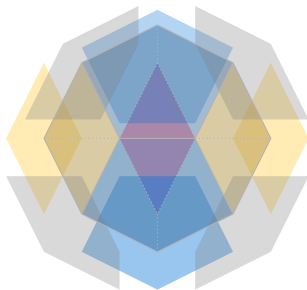
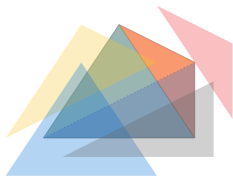
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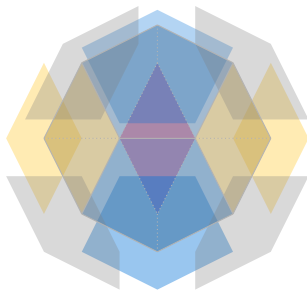
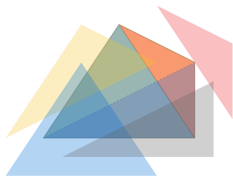
Lower dimensional
SIMPLICES

$$\Delta_2 \hookrightarrow \Delta_3$$

Operadic combinations of
Lower dimensional
ASSOCIAHEDRAS

$$\text{pentagon}_3 \times (\text{pentagon}_3 \times \text{pentagon}_1 \times \text{pentagon}_1) \hookrightarrow \text{pentagon}_5$$

CAN WE BUILD A NERVE OUT OF THE ASSOCIAHEDRA?



FACES ARE

Lower dimensional
SIMPLICES

$$\Delta_{n-1} \hookrightarrow \Delta_n$$

Operadic combinations of
Lower dimensional
ASSOCIAHEDRAS

$$\text{pentagon}_r \times \prod_{i=1}^r \text{pentagon}_{n_i} \hookrightarrow \text{pentagon}_{n_1+\dots+n_r}$$

UNDERSTANDING THE STRUCTURE OF THE REPRESENTABLES

$$\mathbb{P}^1 \hookrightarrow \left[\mathbb{P}^1{}^{\mathrm{op}} \rightarrow \mathbf{SET} \right] \text{ in } \mathbf{Op}_{\mathbf{CAT}}^{\mathbb{N}}$$

We need to work within an *operadic framework* to obtain a relevant notion of presheaf here

UNDERSTANDING THE STRUCTURE OF THE REPRESENTABLES

$$\mathbb{O} \hookrightarrow \left[\mathbb{O}^{\text{op}} \rightarrow \mathbf{SET} \right] \text{ in } \mathbf{Op}_{\mathbf{CAT}}^{\mathbb{N}}$$

We need to work within an *operadic framework* to obtain a relevant notion of presheaf here: suppose a sequence of unrelated presheaves

$$n \in \mathbb{N} \Rightarrow X(n) : \mathbb{O}(n)^{\text{op}} \rightarrow \mathbf{SET} \quad \text{in } \mathbf{CAT}$$

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We can describe this sequence as a single morphism

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$$\mathbf{\heartsuit} \hookrightarrow \left[\mathbf{\heartsuit}^{\mathrm{op}} \rightarrow \mathbf{SET} \right] \text{ in } \mathbf{Op}_{\mathbf{CAT}}^{\mathbb{N}}$$

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We can describe this sequence as a single morphism

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once the appropriate structure is defined on the 2-category $\mathbf{CAT}^{\mathbb{N}}$.

The geometric situation suggests a substitution structure of the form

$$r \in \mathbb{N}, T \in \mathbf{\heartsuit}(r); n_1, \dots, n_r \in \mathbb{N}, T_1 \in \mathbf{\heartsuit}(n_1), \dots, T_r \in \mathbf{\heartsuit}(n_r)$$

$$X(r)_T \times \prod_{i=1}^r X(n_i)_{T_i} \rightarrow X(\Sigma_{i=1}^r n_i)_{T \ast (T_1, \dots, T_r)} \quad \text{in } \mathbf{SET}$$

BACK TO GEOMETRY: PRESHEAF OBJECTS

The presheaves we're interested in consist of operads X internal to the presheaf operad of the face poset of the associahedra

$$X \in \text{MON}_{(\text{CAT}^{\mathbb{N}}, \circ)} \left([\mathbf{\hat{\square}}^{\text{op}}, \mathbf{SET}], \star_{\text{DAY}} \right) := \text{OP}_{\mathbf{SET}}^{\mathbf{\hat{\square}}}$$

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$$X \star_{\text{DAY}} X \rightarrow X \quad \text{in } [\Box^{\text{op}}, \mathbf{SET}]$$

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consists of functors

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meaning that

$$\begin{aligned} r \in \mathbb{N}, \quad n_{\bullet} \in \mathbb{N}^r, \quad & \text{in SET} \\ T \in \mathbf{Op}(r), \quad T_{\bullet} \in \mathbf{Op}^{\bar{\otimes}^r}(n_{\bullet}) \quad \Rightarrow \quad & X(r)_T \times \prod_{i=1}^r X(n_i)_{T_i} \rightarrow X(\Sigma n_{\bullet})_{T \star (T_1, \dots, T_r)} \end{aligned}$$

BACK TO GEOMETRY: REALISATION $\dashv$ NERVE

We obtain an internal adjunction

$$|-|_{\square} : [\square^{\text{op}}, \mathbf{SET}] \rightleftarrows \mathbf{TOP} : N_{\square} \quad \text{in } \mathbf{OP}_{\mathbf{CAT}}^{\mathbb{N}}$$

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that restricts to an adjunction

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The geometric realisation of $X : \mathbf{OP}_{\mathbf{SET}}^{\square}$ is explicitly given by

$$|X|_{\square} \simeq \int_{T \in \square} X_T \cdot \square_T \quad \text{in } \mathbf{OP}_{\mathbf{TOP}}^{\mathbb{N}}$$

The nerve of a topological operad $Y : \mathbf{OP}_{\mathbf{TOP}}^{\mathbb{N}}$ is explicitly given by

$$N_{\square} Y_{\bullet} \simeq \mathbf{TOP}(\square_{\bullet}, Y) \quad \text{in } \mathbf{OP}_{\mathbf{SET}}^{\square}$$

BACK TO GEOMETRY: REALISATIONS OF REPRESENTABLES

So, we have

$$|-|_{\square} : \mathbf{OP}_{\mathbf{SET}}^{\square} \rightleftharpoons \mathbf{OP}_{\mathbf{TOP}}^{\mathbb{N}} : N_{\square} \quad \text{in CAT}$$

BACK TO GEOMETRY: REALISATIONS OF REPRESENTABLES

So, we have

$$|-|_{\square} : \mathbf{OP}_{\mathbf{SET}}^{\square} \rightleftharpoons \mathbf{OP}_{\mathbf{TOP}}^{\mathbb{N}} : N_{\square} \quad \text{in CAT}$$

We also form

$$C_{\square} : \mathbf{OP}_{\mathbf{SET}}^{\square} \rightleftharpoons \mathbf{OP}_{\mathbf{CHZ}}^{\mathbb{N}} : P_{\square} \quad \text{in CAT}$$

BACK TO GEOMETRY: REALISATIONS OF REPRESENTABLES

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The representables form an operad

$$\square(_, \bullet) \in \mathbf{OP}_{\mathbf{SET}}^{\square}$$

BACK TO GEOMETRY: REALISATIONS OF REPRESENTABLES

So, we have

$$|-|_{\triangleleft} : \mathbf{OP}_{\mathbf{SET}}^{\triangleleft} \rightleftharpoons \mathbf{OP}_{\mathbf{TOP}}^{\mathbb{N}} : N_{\triangleleft} \quad \text{in CAT}$$

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By construction:

$$|\triangleleft(_, \bullet)|_{\triangleleft} \simeq \triangleleft \quad \text{in } \mathbf{OP}_{\mathbf{TOP}}^{\mathbb{N}}$$

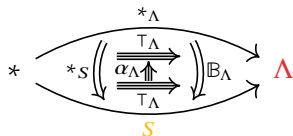
and

$$C_{\triangleleft} \triangleleft(_, \bullet) \simeq A_{\infty} \quad \text{in } \mathbf{OP}_{\mathbf{CHZ}}^{\mathbb{N}}$$

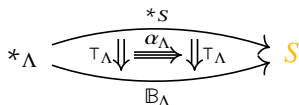
THE FRAMEWORK ENCOMPASSING THIS FRAMEWORK

EVERYTHING IS AN ELEMENT OF SOMETHING ELSE

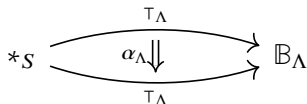
$$\alpha_{\Lambda} : \tau_{\Lambda} : \mathbb{B}_{\Lambda} : \textcolor{brown}{S} : \textcolor{red}{\Lambda} : \text{CAT}_2 : \dots : \text{CAT}_n : \text{CAT}_{n+1} : \dots$$



in CAT_2



in $\textcolor{red}{\Lambda}$



in $\textcolor{brown}{S}$

Thank you very much!

Source:

S. d’Espalungue d’Arros, *Operads in 2-Categories and Models of Structure Interchange* (2023).

<https://theses.hal.science/tel-04617115>.

Contact:

sophiedespalungue@gmail.com



More to come soon... :)